

APPENDIX I

Useful Vector Relations

$$(1) \quad \vec{A} \cdot \vec{B} = \vec{B} \cdot \vec{A} = A_x B_x + A_y B_y + A_z B_z$$

$$(2) \quad \vec{A} \times \vec{B} = -\vec{B} \times \vec{A} = \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ A_x & A_y & A_z \\ B_x & B_y & B_z \end{vmatrix}$$

$$(3) \quad \vec{A} \cdot (\vec{B} \times \vec{C}) = (\vec{A} \times \vec{B}) \cdot \vec{C} = (\vec{C} \times \vec{A}) \cdot \vec{B}$$

$$(4) \quad \vec{A} \times (\vec{B} \times \vec{C}) = (\vec{A} \cdot \vec{C}) \vec{B} - (\vec{A} \cdot \vec{B}) \vec{C}$$

$$(5) \quad (\vec{A} \times \vec{B}) \times \vec{C} = (\vec{A} \cdot \vec{C}) \vec{B} - (\vec{B} \cdot \vec{C}) \vec{A}$$

$$(6) \quad (\vec{A} \times \vec{B}) \cdot (\vec{C} \times \vec{D}) = (\vec{A} \cdot \vec{C}) (\vec{B} \cdot \vec{D}) - (\vec{A} \cdot \vec{D}) (\vec{B} \cdot \vec{C})$$

$$(7) \quad (\vec{A} \times \vec{B}) \times (\vec{C} \times \vec{D}) = [\vec{A} \cdot (\vec{B} \times \vec{D})] \vec{C} - [\vec{A} \cdot (\vec{B} \times \vec{C})] \vec{D}$$

$$(8) \quad \vec{\nabla}(\phi\psi) = \phi \vec{\nabla}\psi + \psi \vec{\nabla}\phi$$

$$(9) \quad \vec{\nabla} \cdot (\phi \vec{A}) = \phi \vec{\nabla} \cdot \vec{A} + \vec{A} \cdot \vec{\nabla}\phi$$

$$(10) \quad \vec{\nabla} \times (\phi \vec{A}) = \phi \vec{\nabla} \times \vec{A} + (\vec{\nabla}\phi) \times \vec{A}$$

$$(11) \quad \vec{\nabla} \cdot (\vec{A} \times \vec{B}) = \vec{B} \cdot (\vec{\nabla} \times \vec{A}) - \vec{A} \cdot (\vec{\nabla} \times \vec{B})$$

$$(12) \quad \vec{\nabla} (\vec{A} \cdot \vec{B}) = (\vec{A} \cdot \vec{\nabla}) \vec{B} + (\vec{B} \cdot \vec{\nabla}) \vec{A} + \vec{A} \times (\vec{\nabla} \times \vec{B}) + \vec{B} \times (\vec{\nabla} \times \vec{A})$$

$$(13) \quad \vec{\nabla} \times (\vec{A} \times \vec{B}) = \vec{A}(\vec{\nabla} \cdot \vec{B}) + (\vec{B} \cdot \vec{\nabla}) \vec{A} - \vec{B}(\vec{\nabla} \cdot \vec{A}) - (\vec{A} \cdot \vec{\nabla}) \vec{B}$$

$$(14) \quad \vec{\nabla} \times (\vec{\nabla} \times \vec{A}) = \vec{\nabla}(\vec{\nabla} \cdot \vec{A}) - (\vec{\nabla} \cdot \vec{\nabla}) \vec{A}$$

$$(15) \quad \vec{\nabla} \cdot (\vec{\nabla} \times \vec{A}) = 0$$

$$(16) \quad \vec{\nabla} \times (\vec{\nabla}\phi) = 0$$

$$(17) (\vec{\nabla} \cdot \vec{\nabla}) \phi = \nabla^2 \phi$$

If $\vec{r}$ is the radius vector, of magnitude r , drawn from the origin to a general point x, y, z , then

$$(18) \vec{\nabla} \cdot \vec{r} = 3$$

$$(19) \vec{\nabla} \times \vec{r} = 0$$

$$(20) \vec{\nabla} r = \vec{r}/r$$

$$(21) \vec{\nabla}(1/r) = -\vec{r}/r^3$$

$$(22) \vec{\nabla} \cdot (\vec{r}/r^3) = -\nabla^2(1/r) = 4\pi \delta(\vec{r})$$

In the following integral relations, V is the volume bounded by the closed surface S and $\hat{n}$ is a unit normal vector drawn outwardly to the closed surface S :

$$(23) \oint_S \phi \hat{n} dS = \int_V (\vec{\nabla} \phi) dV$$

$$(24) \oint_S \vec{A} \cdot \hat{n} dS = \int_V (\vec{\nabla} \cdot \vec{A}) dV \quad (\text{Gauss' theorem})$$

$$(25) \oint_S (\hat{n} \times \vec{A}) dS = \int_V (\vec{\nabla} \times \vec{A}) dV$$

$$(26) \oint_S \phi (\vec{\nabla} \psi) \cdot \hat{n} dS = \int_V [\phi \nabla^2 \psi + (\vec{\nabla} \phi) \cdot (\vec{\nabla} \psi)] dV \quad (\text{Green's first identity})$$

$$(27) \oint_S (\phi \vec{\nabla} \psi - \psi \vec{\nabla} \phi) \cdot \hat{n} dS = \int_V (\phi \nabla^2 \psi - \psi \nabla^2 \phi) dV \quad (\text{Green's second identity or Green's theorem})$$

$$(28) \oint_S [\vec{B} \times (\vec{\nabla} \times \vec{A}) - \vec{A} \times (\vec{\nabla} \times \vec{B})] \cdot \hat{n} dS =$$

$$= \int_V \{ \vec{A} \cdot [\vec{\nabla} \times (\vec{\nabla} \times \vec{B})] - \vec{B} \cdot [\vec{\nabla} \times (\vec{\nabla} \times \vec{A})] \} dV \quad (\text{Vector version of Green's theorem})$$

If S is an open surface bounded by the contour C , of which the line element is $d\vec{\ell}$, then

$$(29) \oint_C \phi \, d\vec{\ell} = \int_S \hat{n} \times (\vec{\nabla} \phi) \, dS$$

$$(30) \oint_C \vec{A} \cdot d\vec{\ell} = \int_S (\vec{\nabla} \times \vec{A}) \cdot \hat{n} \, dS \quad (\text{Stoke's theorem})$$

If $\vec{T}$ is a tensor, then

$$(31) \vec{\nabla} \cdot (\phi \vec{T}) = \phi (\vec{\nabla} \cdot \vec{T}) + (\vec{\nabla} \phi) \cdot \vec{T}$$

$$(32) \oint_S \vec{T} \cdot \hat{n} \, dS = \int_V (\vec{\nabla} \cdot \vec{T}) \, dV$$